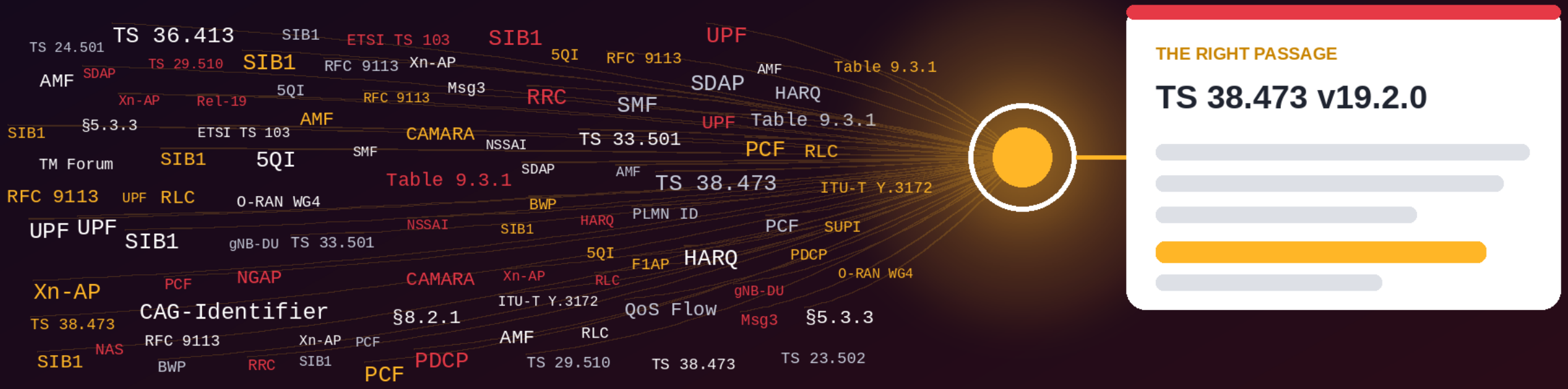
TELCORETRIEVE CHALLENGE

LLMs don’t speak 3GPP.

PRIZE POOL
$25,000
in total prizes

Liam Düro

UCL, Machine Learning Researcher



GSMA OPEN TELCO AI

TelcoRetrieve

One layer of telco knowledge every AI model can reason over it.

ONE LAYER

Telco knowledge in one retrieval layer

Open and vendor-neutral, so every model can reason over telecom standards.

CROSS-SDO RETRIEVAL

One index across many standards bodies

3GPP, ITU-T, IETF, IEEE, ETSI and more in one pipeline.

Your task: make LLMs read 3GPP

TelcoRetrieve is the open retrieval layer. The challenge is to make it better.

THE TASK

1 track

Improve the pipeline

Change how context is retrieved and assembled, or fine-tune the models. Everyone works on the same 3GPP Rel-19 corpus.

HOW IT IS JUDGED

70 / 30

Accuracy and token cost

70% answer accuracy, 30% token efficiency. A leaner prompt scores higher. Minimum accuracy 80%.

WHAT YOU WIN

$25K

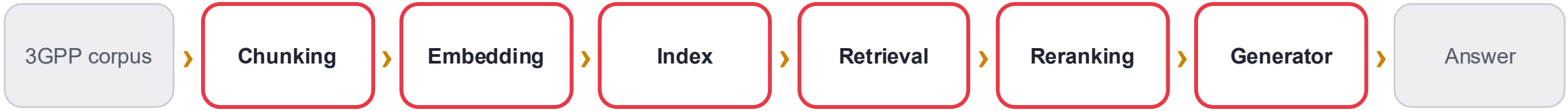
Prizes and MWC Barcelona

Cash for the top five. Top three also get an MWC Barcelona Leader Pass and travel. Winners announced in March 2027.

13 NOV – 13 JAN **100 public questions from the start · 200 private questions on 8 Jan · top 10 hand over their code**

TELCORETRIEVE CHALLENGE · THE TRACK

One track: Pipeline and Model Optimization



Red steps are yours to change. Grey steps are fixed.

WHAT YOU CAN CHANGE

Chunking and index	Strategy, size, overlap
Retrieval and reranking	method, how many passages
The prompt	how context is assembled
The models	fine-tune the embedder or Gemma 3n

WHAT STAYS FIXED

3GPP Rel-19 only
the corpus is provided

OTel-Reranker-0.6B
the reranker model

4,096 tokens
maximum prompt length

SUBMIT your answers and input prompts · **TOP 10** hand over the whole pipeline to be checked

* Embedder: OTel-Embedding-0.6B (farbodtavakkoli/OTel-Embedding-0.6B) · Reranker: OTel-Reranker-0.6B (farbodtavakkoli/OTel-Reranker-0.6B) · Base model: Gemma 3n E4B instruct (google/gemma-3n-e4b-it)

TELCORETRIEVE CHALLENGE · SUBMISSION

What you receive and what you submit

YOU RECEIVE

Multiple choice | 3GPP Rel-19 | TS 38.473 v19.2.0

According to the specification, what is the maximum number of CAG-Identifiers that can be broadcast in a cell? (according to 3GPP Rel-19, TS 38.473 v19.2.0)

- **A** 16
- **B** 12
- **C** 10

also true / false

YOU SUBMIT

One CSV, header `ID,input_prompt,answer`

1. `ID` — Copied unchanged. Keep it as text.
2. `input_prompt` — What Gemma sees. Max 4,096 tokens.
3. `answer` — One letter A–E, or TRUE / FALSE.

```
"123456789012","You are an expert on 3GPP… Question: …","B"
```

HOW IT RUNS

13 NOV

100 public questions

Submit up to 300 times. The public leaderboard updates after every submission.

8 JAN

+200 private questions

Released before the close. Submit these too. Scores stay hidden until the end.

13 JAN

Top 10 hand over code

Your whole pipeline is checked by Zindi, then the final ranking is published.

How submissions are scored

PUBLIC LEADERBOARD

100 public questions Live from 13 Nov · scores shown after every submission

PRIVATE LEADERBOARD · HIDDEN UNTIL THE END

200 private questions Released 8 Jan · scores revealed only after the close

ACCURACY

Multiple-choice and true/false

Exact match with the gold answer. One option letter or TRUE / FALSE only; anything else scores 0.

Accuracy = 100 × correct / questions

CODE

`accuracy.py`

prints accuracy, 0–100

TOKEN EFFICIENCY

Tokens in your input_prompt

Counted over your input_prompts. A prompt over 4,096 tokens rejects the whole submission.

Token efficiency = 100 × B / (B + T)

B = 850 × questions; T = your tokens

CODE

`token_efficiency.py`

tiktoken cl100k_base

FINAL SCORE

One mixed metric

Fewer tokens raise the score; wrong answers lower it.

Score out of 100. Minimum accuracy: 80%

0.7 × Accuracy + 0.3 × Token efficiency

CODE

`overall_score.py`

prints the final score, 0–100

LEADERBOARD **Public ranking uses the 100 public questions. The final ranking uses the 200 private questions.**

TOP 10 **Hand over your whole pipeline (code and configuration) to Zindi to be checked.**

Scoring, a worked example

Same 100 questions, three pipelines. Baseline B = 850 × 100 = 85,000 tokens. Illustrative numbers.

PIPELINE A

Retrieve a lot

Accuracy	**90%**
Tokens (T)	**85,000**
	850 tokens per question
Token efficiency	**50.0**

FINAL SCORE
78.0

PIPELINE B

Trim the prompt

Accuracy	**85%**
Tokens (T)	**50,000**
	500 tokens per question
Token efficiency	**63.0**

FINAL SCORE
78.4

PIPELINE C

One tight chunk

Accuracy	**82%**
Tokens (T)	**25,000**
	250 tokens per question
Token efficiency	**77.3**

FINAL SCORE
80.6

Accuracy counts for 70%, token efficiency for 30%. A leaner prompt can beat a more accurate pipeline: C wins with 8 points less accuracy than A.

RULES Minimum accuracy 80% · every input_prompt at most 4,096 tokens (one longer prompt rejects the whole submission) · tokens counted with tiktoken cl100k_base

What not to do

DON'T Leave rows out

```
99 rows for 100 questions
```

Result `Missing entries for IDs …`

Do One row for every question ID in the file.

DON'T Repeat an ID

```
"09436ccaeebf",… (two rows)
```

Result `Duplicate IDs …`

Do Each ID exactly once.

DON'T Send an empty prompt

```
"09436ccaeebf","","B"
```

Result `Empty input_prompt for IDs …`

Do The full prompt Gemma saw, never blank.

DON'T Go over 4,096 tokens

```
input_prompt = 4,500 tokens
```

Result `input_prompt over 4096 tokens for IDs …`

The whole submission is rejected.

Do Trim passages. Check with token_efficiency.py.

DON'T Answer in sentences

```
"The answer is B" "b) 12"
```

Result Cannot be read, so it scores 0.

Do B or TRUE / FALSE. Nothing else.

DON'T Let Excel touch the IDs

```
123456789012 → 1.23457E+11
```

Result `Missing entries for IDs …`

Do Never open or save the CSV in Excel. Read IDs as text.

ALSO Header exactly **`ID,input_prompt,answer`**: no extra columns, no missing header.

CHECK FIRST Run accuracy.py, token_efficiency.py and overall_score.py on your own CSV

From announcement to awards

10 Nov 2026	13 Nov	8 Jan	13 Jan	15 Jan	End Jan	Mar 2027
Announcement	**Competition opens**	**200 new questions**	**Competition closes**	**Top 10 selected**	**Final leaderboard**	**Awards**
Challenge announced at MWC Doha	Toolkit and 100 public questions. Public leaderboard goes live	Private questions released. Submit these too	Last day to submit	Hand over your code and configuration for checking	Code Review and final results	Awards at MWC Barcelona

PUBLIC LEADERBOARD
100 questions, live from 13 Nov

PRIVATE LEADERBOARD
200 questions from 8 Jan, hidden until the end

TELCORETRIEVE CHALLENGE · RESOURCES

Everything you need to build

PRIZE POOL: €25K



Toolkit Znbne/telco-retrieve
The RAG pipeline itself: ingest, search, rerank and answer from the command line.
GITHUB

Indexed data
Ready-made Rel-19 chunks and index plus the question data. Start tuning without re-ingesting.
HUGGING FACE

Guidebook Znbne/telcoretrieve-cookbook
Worked examples for each stage: chunking, retrieval, reranking and generation.
GITHUB

Raw dataset GSMA/3GPP
The 3GPP specifications as text, if you want to build your own chunks.
HUGGING FACE

Training data telecom-standards-corpus
Telecom standards text for fine-tuning your embedder or Gemma 3n.
HUGGING FACE

Scoring scripts Znbne/telco-retrieve
accuracy.py, token_efficiency.py and overall_score.py: check your CSV before you submit.
GITHUB

Prizes and where to start

PRIZE POOL: $25K

PRIZES (USD)

1st	$8,000 USD
2nd	$5,000 USD
3rd	$3,500 USD
4th	$2,000 USD
5th	$1,500 USD

+ MWC Barcelona Leader Pass (worth $2,000)
+ up to $1,500 USD in costs to attend MWC Barcelona in March 2027 (one representative)

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+ up to $1,500 USD in costs to attend MWC Barcelona in March 2027 (one representative)

Plus 10,000 Zindi points available.

START HERE

github.com/Znbne/telco-retrieve

Competition opens 13 Nov 2026. Closes 13 Jan 2027.



MWC BARCELONA

Winners are announced at MWC Barcelona, March 2027

Join the challenge



Search: TelcoRetrieve Challenge on Zindi

zindi.world/competitions/telcoretrieve-challenge-unlocking-telco-knowledge

Registration closes 6 Jan 2027 · teams of up to 3